Check for updates

Can Human Growth Hormone Accelerate Tendon and Ligament Injury Recovery?

Keith M. Baumgarten, MD,[†‡*] Eric S. Sandhurst, PhD,[§] and Hongli Sun, PhD[§]

Background: Studies involving human fibroblasts and use of human growth hormone (HGH) administration for injury recovery are limited. It is plausible that if the administration of HGH to human cells increased cellular proliferation and differentiation, then HGH might be able to assist in accelerating recovery from injury.

Hypothesis: HGH will increase proliferation and differentiation of human tendon and ligament fibroblasts in vitro based on both a single-dose and a sustained-dose model of HGH administration.

Study Design: Basic science cellular study.

Methods: Human tendon and ligament tissue were harvested from 24 patients. Tissue samples were digested with type I collagenase to isolate the target cell types. HGH was administered directly to isolated cells at doses ranging from 100 pg/mL to 10 µg/mL, either in a single-dose or a sustained-dose model. Proliferation was analyzed at days 4 and 7. Differentiation of ligament and tendon fibroblasts was assessed at day 14.

Results: Administration of a single-dose of HGH to both cell types demonstrated similar or inferior cellular proliferation compared with controls after 7 days. For the sustained-dosing model of ligament fibroblasts, only the 100 ng/mL concentration demonstrated at least statistically similar or improved proliferation compared with controls. When examining the 100 ng/mL HGH concentration with larger sample sizes, cellular proliferation was not improved over controls for any cell type for the single- or sustained-dosing models. Proliferation for tendon fibroblasts was either similar or inferior to the control group at all concentrations of HGH. There was no clear dose-response relationship demonstrating enhanced collagen production with administration of HGH to suggest it enhances injury recovery.

Conclusion: HGH administered to human tendon and ligament fibroblasts does not appear to positively affect cellular proliferation and differentiation.

Clinical Relevance: This study does not support the use of HGH for accelerating recovery from injury.

Keywords: cellular proliferation; fibroblasts; human growth hormone; ligament; tendon

Human growth hormone (HGH) has been approved by the United States Food and Drug Administration (FDA) for the treatment of pediatric growth hormone deficiency, adult growth hormone deficiency, short stature, and AIDS-related muscle wasting.[16] Although it is not FDA-approved to enhance recovery after injuries, athletes have been using HGH to accelerate recovery time from a variety of injuries.[24] Although anecdotal, many athletes believe that the use of these mediators have facilitated an early return to competition after injury.[24]

There are few scientific studies that have examined the role of HGH as an independent therapeutic to accelerate recovery time from injury.[1,4,6,22,28] Specifically, studies that examine the effects of HGH on ligament or tendon fibroblast proliferation are rare.

From [†]Orthopedic Institute, Sioux Falls, South Dakota, [‡]Sanford School of Medicine, University of South Dakota, Vermillion, South Dakota, and [§]University of South Dakota GEAR Center, Sioux Falls, South Dakota

*Address correspondence to Keith M. Baumgarten, MD, Orthopedic Institute, 810 E 23rd Street, Sioux Falls, South Dakota 57117 (email: kbaumga@yahoo.com) (Twitter: @DrBaumgarten).

The following author declared potential conflicts of interest: KMB received consulting fees and speaking payments from Miach and Stryker.

This study was funded by the ST Li foundation and Pfizer donated the human growth hormone utilized in this study. The study sponsors did not have any influence on study design, data collection, data interpretation, writing of the report, nor in the decision for publication of this manuscript.

DOI: 10.1177/19417381241245938

© 2024 The Author(s)

We could identify only 1 study using rat periodontal ligament fibroblasts, which did not demonstrate an effect of HGH on cellular growth.[7]

In the current study, the effects of HGH administered directly to human ligament and tendon fibroblasts were examined to determine whether it would increase cellular proliferation and differentiation which might enhance injury recovery. The aims of this study were (1) to determine whether HGH can accelerate or improve ligament and tendon cellular proliferation compared with controls, (2) to determine a dose-response relationship between HGH and ligament and tendon cellular proliferation, (3) to assess whether a single or sustained dose of HGH administration affected ligament and tendon cellular proliferation compared with controls, and (4) to determine whether HGH can increase collagen formation compared with controls. The hypothesis of this study was that HGH would increase cellular proliferation and increase markers of injury recovery (collagen) in human tendon and ligament fibroblast cell cultures.

METHODS

Once institutional review board approval was obtained, ligament fibroblasts were harvested from the anterior cruciate ligament (ACL) during total knee arthroplasty (13 patients) and tendon fibroblasts from patients undergoing ACL reconstruction utilizing a hamstring tendon autograft (11 patients). The excess tendon that was available after ligament reconstruction, which otherwise would be discarded, was utilized for cell culture.

This study utilized a range of HGH dosing (100 pg/mL to 100 µg/mL) that spanned subphysiologic, physiologic, and even supraphysiologic levels, and incorporated the spectrum of concentrations that have been documented in previous cell culture studies.[12,13,15,17,19,25]

To model the effects of a single 1-time injection, commercially available HGH (Genotropin, Pfizer) was administered into the cell culture medium only once at the initiation of the experiment. To model a sustained-release delivery, HGH was administered into the cell culture medium at the beginning of the experiment and when the culture medium was replenished at days 2, 4, and 6.

All experiments were carried out in 24-well culture plates (Falcon, Corning) with 3 negative control wells containing tissue-derived cells, medium, and MTS (3-(4,5-dimethylthiazol-2-yl)-5-(3-carboxymethoxyphenyl)-2-(4-sulfophenyl)-2H-tetrazolium) assay solution and 3 wells containing tissue-derived cells, medium, MTS assay solution, and various experimental doses of HGH. Follow-up experiments for any dosages that revealed potentially beneficial effects on cellular proliferation or differentiation were examined further using larger sample sizes. This was carried out with 6 control wells containing tissue-derived cells, medium, and MTS assay solution, and 18 experimental wells containing tissue-derived cells, HGH, medium, and MTS assay solution to determine the effects of the dosage(s) of HGH that revealed potentially beneficial effects with the initial lower sample size experiment.

Cell Isolation and Culture

Tissue specimens were rinsed with phosphate-buffered saline (PBS) solution (Gibco) then dissected into small pieces (~1 mm^3). Ligament and tendon tissues were digested in 0.3% collagenase type I (Gibco) in Hanks Buffered Salt Solution (HBSS) (Gibco) for 200 minutes under constant shaking at 37°C.[15,18,32,33] The digested tissue was passed through a 40-µm nylon mesh cell strainer (Falcon, Corning) and any remaining solid material was discarded. The flowthrough was centrifuged (Eppendorf) at 1500 rpm for 5 minutes. The supernatant was aspirated and discarded, and the cells were resuspended in Dulbecco's Modified Eagle Medium (DMEM) containing 10% fetal bovine serum (FBS) (Gibco) plus 1% penicillin (10,000 units/mL)-streptomycin (10,000 µg/mL) (Gibco) to stop the digestion. The suspended cells were placed on treated tissue culture dishes (Falcon, Corning) and incubated at 37°C in a humidified atmosphere of 5% CO_2 and 95% air. After 24 hours, nonadherent cells were removed by rinsing twice with PBS solution. When cells reached near-confluence, they were detached from the culture dish using 0.25% Trypsin (Gibco), replated into 24-well culture dishes (Falcon, Corning), and used no later than passage 3.

Cell Proliferation

The cell culture methodology described here has been used widely for human cellular studies.[23,27,30-33] For both single- and sustained-dose cell proliferation experiments, isolated cells were plated at a density of 5000 cells per well in 24-well plates. The control groups received only DMEM with 10% FBS and 1% penicillin-streptomycin. The experimental groups received 500 µl of DMEM with 10% FBS and 1% penicillin-streptomycin combined with various doses of HGH ranging from 100 pg/mL to 100 µg/mL. The effects of the various doses of HGH were studied in triplicate. Cell proliferation was assessed by using the CellTiter 96 Aqueous One Solution Cell Proliferation Assay (MTS) (Promega). At the end of the culture period, 50 µL MTS assay solution was added to each well containing 450 µL of fresh culture medium. The plate was then kept in a 37°C, 5% CO_2 incubator for 1 hour before aliquoting 400 µL of the reacted solution per each well of the 24-well plate. Once the reacted medium was aliquoted in quadruplicate (100 µL into each of 4 wells of a 96-well plate), the optical density (OD), or absorbance, was measured at 490 nm using a microplate reader (Tecan) and the resulting value was correlated to the number of cells.

Cell Differentiation

To study the effect of HGH on ligament fibroblast and tendon fibroblast phenotype, we measured collagen production at 14 days relative to the untreated control group.[10] Culture medium was removed and cells were rinsed twice with PBS solution to remove residual medium. The cells were fixed with 3.7% formaldehyde (Ricca Chemical Company) for 10 minutes at room temperature. The wells were washed twice with PBS solution to remove residual formaldehyde. Picrosirius Red (1 mL

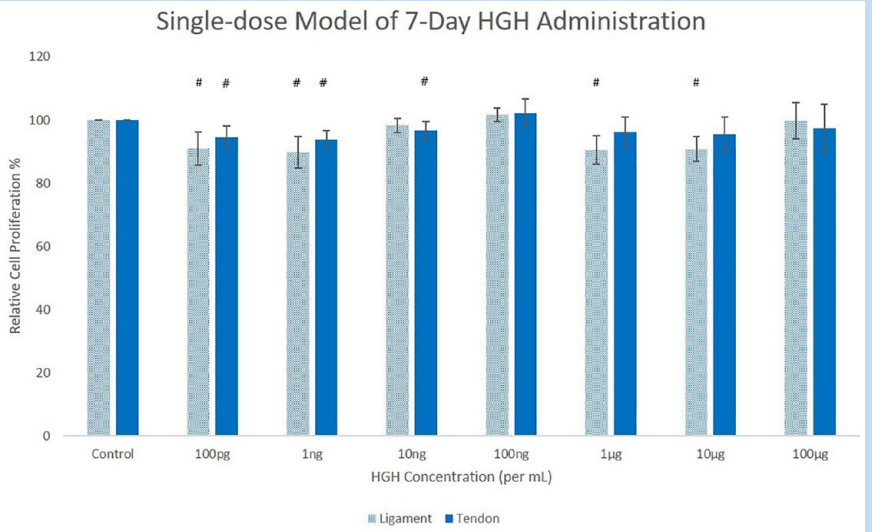

Figure 1. Relative cell proliferation of ligament and tendon fibroblasts after administration of a single dose of HGH compared with untreated control groups. Data are expressed as means and standard deviations. #Significant decreases in proliferation compared with control ($P \leq 0.05$). HGH, human growth hormone.

of 0.1% solution; Sigma) stain was added to each well and the plates were placed on an orbital shaker (Talboys) rotating at 100 rpm for 18 to 24 hours. The excess Picrosirius Red solution was removed from the wells and each well was rinsed 4 times with deionized water. Then, 500 µL of a 50:50 ratio of methanol (Fisher Scientific) to 0.1 M NaOH (Acros Organics) was added to each well and left at room temperature for 30 minutes. The solution was transferred to a 96-well plate and each well was analyzed in quadruplicate by removing 100 µL and adding it to each of 4 wells. The absorbance was analyzed using a Tecan Microplate Reader set for 490 nm. Results were compared with the untreated groups.

STATISTICAL ANALYSIS

Each experimental group had at least 3 biological replicates, and each biological replicate was analyzed in triplicate as 3 technical replicates. Furthermore, each technical replicate was analyzed in quadruplicate by dividing the MTS assay supernatant into 4 groups for a total sample size of 36 per group. To achieve a power of at least 0.8, a minimum sample size of 26 would be required to achieve a significance level of 0.05 and an effect size of 80%. To determine statistical significance of the OD/absorbance values for each experimental group receiving a different dose of HGH, results were compared individually with the control using a 1-way ANOVA. A value of $P \leq 0.05$ was considered to be statistically significant.

RESULTS

The administration of a single-dose of HGH to both cell types demonstrated either similar or inferior cellular proliferation compared with controls after 7 days (Figure 1). Ligament fibroblasts demonstrated significantly reduced proliferation at 100 pg/mL (90.9%, $P < 0.01$), 1 ng/mL (89.7%, $P < 0.01$), 1 µg/mL (90.4%, $P < 0.01$), and 10 µg/mL (90.7%, $P < 0.01$) concentrations. Tendon fibroblasts displayed significantly reduced proliferation at doses of 100 pg/mL (94.5%, $P < 0.01$), 1 ng/mL (93.7%, $P < 0.01$), and 10 ng/mL (96.7%, $P = 0.03$).

For the sustained-dose model at the 4-day timepoint, cellular proliferation for tendon fibroblasts was either similar or inferior to the control group at all concentrations of HGH (Figure 2). Cellular proliferation in tendon fibroblasts was statistically significantly decreased at the 4-day timepoint at 1 µg/mL (97.41%, $P = 0.05$) and 100 µg/mL (88.5%, $P = 0.01$) concentrations compared with controls. However, in the sustained-dose model for ligament fibroblasts at the 4-day timepoint, 2 of the 7 concentrations studied (100 ng/mL [107.7%, $P = 0.03$] and 10 µg/mL [104.6%, $P < 0.01$]) exhibited increased cellular proliferation compared with controls, with the remainder of the concentrations demonstrating similar proliferation to controls. None of the concentrations at the 7-day timepoint for ligament or tendon fibroblasts demonstrated statistically improved cellular proliferation, with the 10 µg dose demonstrating significantly inferior cellular proliferation in tendon fibroblasts compared with controls.

Since the 100 ng/mL HGH concentration was the only concentration of HGH that demonstrated either similar or increased cellular proliferation compared with controls for both cell types with both single-dose and sustained-dose models, this concentration was analyzed further with larger sample sizes to determine whether it would enhance cellular proliferation. When examining the 100 ng/mL HGH concentration with larger sample sizes (n = 18), cellular proliferation was not improved

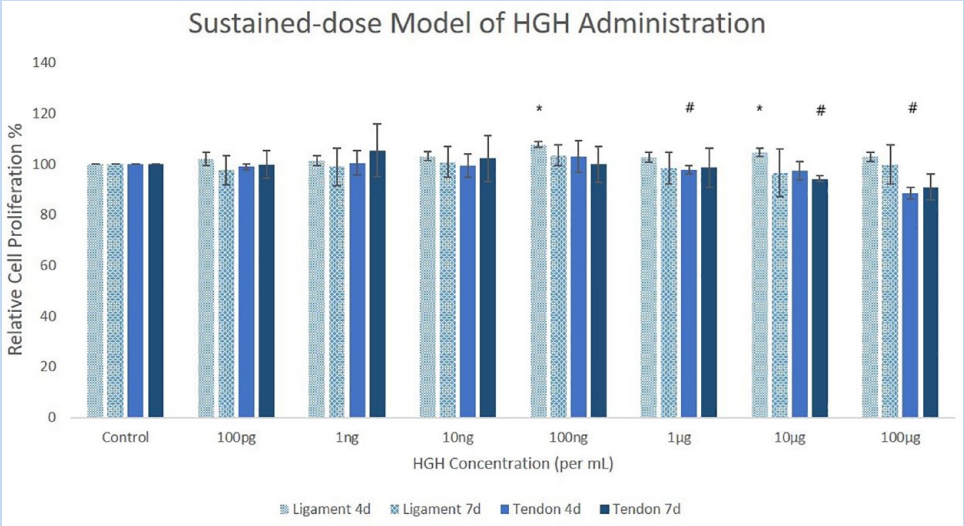

Figure 2. Relative cell proliferation of ligament and tendon fibroblasts after administration of sustained-dose of HGH compared with untreated control groups. Data are expressed as means and standard deviations. #Significant decreases in proliferation compared with control; *significant increases in proliferation compared with control ($P \leq 0.05$). HGH, human growth hormone.

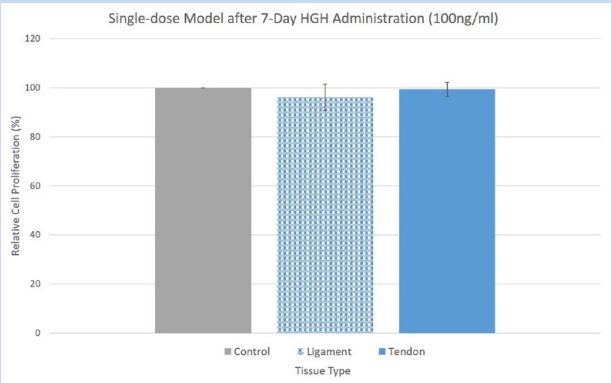

Figure 3. Relative cell proliferation of ligament and tendon fibroblasts after administration of a single 100 ng/mL dose of HGH. No significant increase or decrease was observed compared with untreated control groups ($P \leq 0.05$). HGH, human growth hormone.

over controls for any cell type for the single-dose model (ligament 96.1%, $P = 0.33$; tendon 99.5%, $P = 0.63$). Likewise, the sustained-dose model of 100 ng/mL indicated that ligament and tendon-derived cells experienced either statistically similar or lower proliferation compared with controls (7 day [ligament] = 95.1%, $P < 0.01$; 4 day [tendon] = 95.6%, $P = 0.03$) (Figures 3 and 4).

When examining the effects of HGH on cellular differentiation, collagen production for all HGH concentrations for both cell types was either similar or inferior to controls. HGH significantly decreased collagen production in ligament fibroblasts at the 1 ng/mL dosage (93.5%, $P = 0.05$) (Figure 5).

DISCUSSION

The results of this study did not support our hypothesis. HGH administration did not significantly improve cellular proliferation. No clear dose-dependent relationship was observed between HGH and improved cellular proliferation of ligament and tendon fibroblasts. Neither single-dose nor sustained-dose models of HGH administration demonstrated a clear enhancement of cellular proliferation compared with the control group. HGH did not increase collagen production compared with controls; rather, our results showed similar or lower collagen production across both tissue types and all doses. A noteworthy finding of this study was that, at many concentrations, there were detrimental effects of HGH with statistically significantly decreased cellular proliferation and collagen production compared with controls.

There have been variable results in animal studies that have examined the role of HGH as an accelerant for injury recovery.[13,14,21] Growth hormone treatment has been shown to increase collagen deposition at wound healing sites in the rat model.[11] Rats treated with growth hormone had a 400% increase in ultimate load and maximum stiffness of tibial fractures compared with controls.[4] Other basic science studies have supported these findings.[2,3,5,18,29] On the contrary, a previous study did not reveal any improvements in biomechanical properties of the rat rotator cuff tendon after repair compared with placebo. This study even raised the concern that HGH may cause inferior biomechanical outcomes compared with placebo.[6] Another study examining Achilles tendon healing in the rat did not demonstrate a benefit of HGH.[1]

The results of in vivo human studies vary in their support of HGH in enhancing injury recovery. Two studies demonstrated

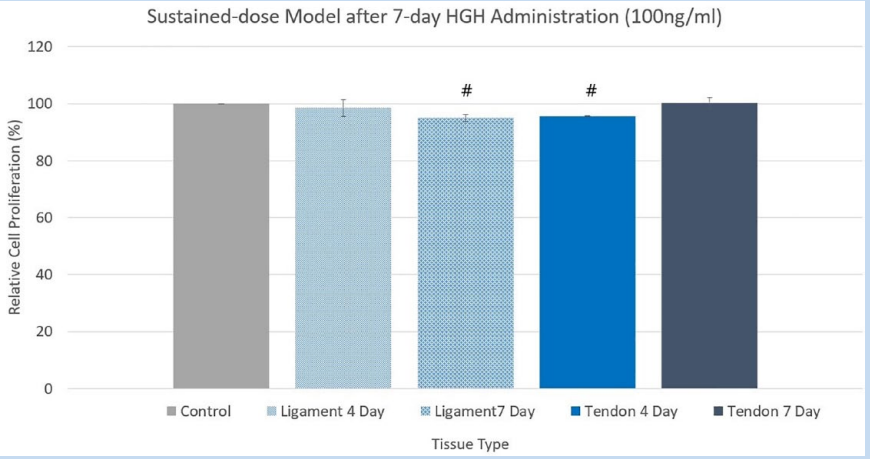

Figure 4. Relative cell proliferation of ligament and tendon fibroblasts after administration of a sustained 100 ng/mL dose of HGH. Data are expressed as means and standard deviations. #Significant decreases in proliferation compared with control ($P \leq 0.05$). HGH, human growth hormone.

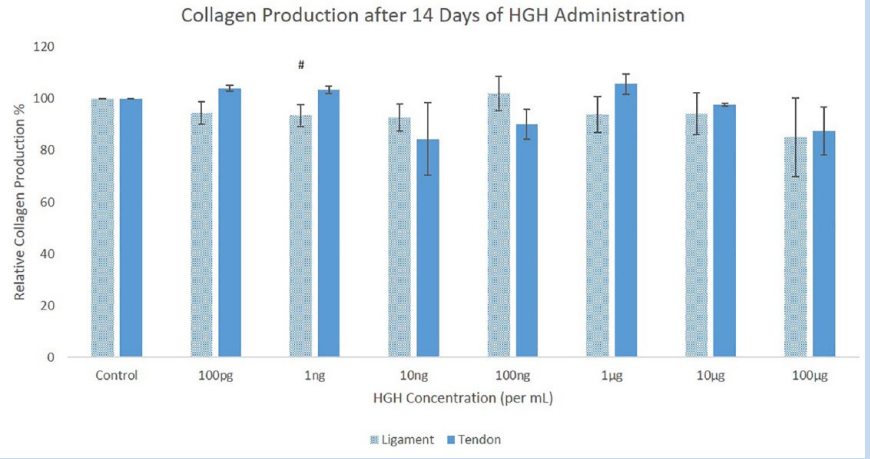

Figure 5. Relative collagen production of ligament and tendon fibroblasts 14 days after sustained administration of HGH. Averages are compared with untreated control groups. Data are expressed as means and standard deviations. #Significant decreases in proliferation compared with control ($P \leq 0.05$). HGH, human growth hormone.

increased collagen expression, tendon cross-sectional area, and tendon stiffness in both young and elderly male patients after limb immobilization and subsequent rehabilitation.[8,9] However, a randomized, double-blinded, placebo-controlled human trial showed no significant enhancement of tibial fracture healing after intramedullary fixation with HGH treatment.[28]

One strength of this study is that it is a human model, which limits the concern of an effect of HGH on a different species. In addition, the dosage of HGH utilized in this study incorporated the spectrum of concentrations that have been documented in previous cell culture studies and spanned human subphysiologic, physiologic, and supraphysiologic levels.[12,13,15,17,19,25] Lastly, this study incorporates human hamstring tenocytes and ACL fibroblasts, which may be more relevant to a study design examining the effects of HGH on injury recovery compared with the human alveolar cells that have been utilized in previous studies.[13] To the best of our knowledge, this is the first published study that examines the effect of HGH on human hamstring tenocytes and ACL fibroblasts.

This current study was limited by the in vitro cellular model, in which cells may not behave as they would in vivo. HGH is released from the pituitary in a pulsatile fashion that is quite difficult to reproduce in both in vitro and in vivo models.[20,26] This study did not try to reproduce the in vivo release of HGH; rather, it attempted to model how HGH might be administered exogenously to a patient. In addition, the effect of HGH was examined in a paracrine fashion. It is entirely possible that when HGH is utilized in an endocrine fashion it may affect the cells in a

different manner. However, it is challenging to perform comprehensive human cellular research and/or tissue biomechanical research that examines the endocrine effects of HGH. A previous animal study that examined the endocrine effect of HGH did not reveal improvements in biomechanical outcomes.[6]

Another limitation of this study is the variable ages of the donor cells. To ethically obtain human cells for this study, we had to identify the rare opportunities in which human cells could be obtained without causing unnecessary harm or risk to human subjects. Tendon fibroblasts were harvested from excess tissue obtained from tendon grafts utilized in patients undergoing ACL reconstruction. These patients range from teenagers to young adults. Ligament fibroblasts were harvested from older patients undergoing arthroplasty procedures. Thus, we were unable to obtain a homogenous sample of human cells. As a result, this study may be confounded by the potential differential response of different age cells to HGH. The concern for this limitation may be lessened since this study did not demonstrate significant differences to the response of each cell type to HGH. The inclusion of variable ages of cells actually increases the generalizability of these results.

CONCLUSION

This study did not demonstrate a dose-dependent trend for increased cellular proliferation or production of collagen that would suggest that HGH administration directly to cells facilitates injury recovery. At best, cellular proliferation and differentiation were similar to controls. In multiple dosing scenarios, cellular proliferation and differentiation were worse after HGH administration. According to our study, exogenous HGH administered in a paracrine fashion to increase cellular proliferation and differentiation in a human cellular model in hopes of accelerating injury recovery is not effective and is potentially detrimental.

REFERENCES

1. Andersson T, Eliasson P, Aspenberg P. Growth hormone does not stimulate early healing in rat tendons. *Int J Sports Med*. 2012;33(3):240-243.
2. Andreassen TT, Oxlund H. Local anabolic effects of growth hormone on intact bone and healing fractures in rats. *Calcif Tissue Int*. 2003;73(3):258-264.
3. Bak B, Jørgensen PH, Andreassen TT. Dose response of growth hormone on fracture healing in the rat. *Acta Orthop Scand*. 1990;61(1):54-57.
4. Bak B, Jørgensen PH, Andreassen TT. Increased mechanical strength of healing rat tibial fractures treated with biosynthetic human growth hormone. *Bone*. 1990;11(4):233-239.
5. Bak B, Jørgensen PH, Andreassen TT. The stimulating effect of growth hormone on fracture healing is dependent on onset and duration of administration. *Clin Orthop Relat Res*. 1991;264:295-301.
6. Baumgarten KM, Oliver HA, Foley J, et al. Human growth hormone may be detrimental when used to accelerate recovery from acute tendon-bone interface injuries. *J Bone Joint Surg Am*. 2013;95(9):783-789.
7. Blom S, Holmstrup P, Dabelsteen E. The effect of insulin-like growth factor-I and human growth hormone on periodontal ligament fibroblast morphology, growth pattern, DNA synthesis, and receptor binding. *J Periodontol*. 1992;63(12):960-968.
8. Boesen AP, Dideriksen K, Couppe C, et al. Effect of growth hormone on aging connective tissue in muscle and tendon: gene expression, morphology, and function following immobilization and rehabilitation. *J Appl Physiol (1985)*. 2014;116(2):192-203.
9. Boesen AP, Dideriksen K, Couppe C, et al. Tendon and skeletal muscle matrix gene expression and functional responses to immobilisation and rehabilitation in young males: effect of growth hormone administration. *J Physiol*. 2013;591(23):6039-6052.
10. Cheng MT, Yang HW, Chen TH, Lee OK. Modulation of proliferation and differentiation of human anterior cruciate ligament-derived stem cells by different growth factors. *Tissue Eng Part A*. 2009;15(12):3979-3989.
11. Christensen H, Oxlund H. Growth hormone increases the collagen deposition rate and breaking strength of left colonic anastomoses in rats. *Surgery*. 1994;116(3):550-556.
12. Cook JJ, Haynes KM, Werther GA. Mitogenic effects of growth hormone in cultured human fibroblasts. Evidence for action via local insulin-like growth factor I production. *J Clin Invest*. 1988;81(1):206-212.
13. Crippa GE, Beloti MM, Cardoso CR, Silva JS, Rosa AL. Effect of growth hormone on in vitro osteogenesis and gene expression of human osteoblastic cells is donor-age-dependent. *J Cell Biochem*. 2008;104(2):369-376.
14. Doessing S, Heinemeier KM, Holm L, et al. Growth hormone stimulates the collagen synthesis in human tendon and skeletal muscle without affecting myofibrillar protein synthesis. *J Physiol*. 2010;588(2):341-351.
15. Haase HR, Ivanovski S, Waters MJ, Bartold PM. Growth hormone regulates osteogenic marker mRNA expression in human periodontal fibroblasts and alveolar bone-derived cells. *J Periodontal Res*. 2003;38(4):366-374.
16. Hintz RL. Growth hormone: uses and abuses. *BMJ*. 2004;328(7445):907-908.
17. Kassem M, Blum W, Ristelli J, Mosekilde L, Eriksen EF. Growth hormone stimulates proliferation and differentiation of normal human osteoblast-like cells in vitro. *Calcif Tissue Int*. 1993;52(3):222-226.
18. Kolbeck S, Bail H, Schmidmaier G, et al. Homologous growth hormone accelerates bone healing - a biomechanical and histological study. *Bone*. 2003;33(4):628-637.
19. Langdahl BL, Kassem M, Møller MK, Eriksen EF. The effects of IGF-I and IGF-II on proliferation and differentiation of human osteoblasts and interactions with growth hormone. *Eur J Clin Invest*. 1998;28(3):176-183.
20. Lee SH, Piao H, Cho YC, et al. Implantable multireservoir device with stimulus-responsive membrane for on-demand and pulsatile delivery of growth hormone. *Proc Natl Acad Sci USA*. 2019;116(24):11664-11672.
21. Leung KC, Waters MJ, Markus I, Baumbach WR, Ho KK. Insulin and insulin-like growth factor-I acutely inhibit surface translocation of growth hormone receptors in osteoblasts: a novel mechanism of growth hormone receptor regulation. *Proc Natl Acad Sci USA*. 1997;94(21):11381-11386.
22. Mendias CL, Enselman ERS, Olszewski AM, et al. The use of recombinant human growth hormone to protect against muscle weakness in patients undergoing anterior cruciate ligament reconstruction: a pilot, randomized placebo-controlled trial. *Am J Sports Med*. 2020;48(8):1916-1928.
23. Miszuk JM, Xu T, Yao Q, et al. Functionalization of PCL-3D electrospun nanofibrous scaffolds for improved BMP2-induced bone formation. *Appl Mater Today*. 2018;10:194-202.
24. Mitchell GJ. Report to the commissioner of baseball of an independent investigation into the illegal use of steroids and other performance enhancing substances by players in Major League Baseball. New York, NY: Office of the Commissioner of Baseball; 2007.
25. Mrak E, Villa I, Lanzi R, Losa M, Guidobono F, Rubinacci A. Growth hormone stimulates osteoprotegerin expression and secretion in human osteoblast-like cells. *J Endocrinol*. 2007;192(3):639-645.
26. Nindl BC, Hymer WC, Deaver DR, Kraemer WJ. Growth hormone pulsatility profile characteristics following acute heavy resistance exercise. *J Appl Physiol (1985)*. 2001;91(1):163-172.
27. Picot J. *Human Cell Culture Protocols*. 2nd ed. Totowa, NJ: Humana Press; 2005.
28. Raschke M, Rasmussen MH, Govender S, Segal D, Suntum M, Christiansen JS. Effects of growth hormone in patients with tibial fracture: a randomised, double-blind, placebo-controlled clinical trial. *Eur J Endocrinol*. 2007;156(3):341-351.
29. Steenfos HH, Jansson JO. Growth hormone stimulates granulation tissue formation and insulin-like growth factor-I gene expression in wound chambers in the rat. *J Endocrinol*. 1992;132(2):293-298.
30. Yao Q, Cosme JG, Xu T, et al. Three dimensional electrospun PCL/PLA blend nanofibrous scaffolds with significantly improved stem cells osteogenic differentiation and cranial bone formation. *Biomaterials*. 2017;115:115-127.
31. Yao Q, Liu Y, Sun H. Heparin-dopamine functionalized graphene foam for sustained release of bone morphogenetic protein-2. *J Tissue Eng Regen Med*. 2018;12(6):1519-1529.
32. Yao Q, Liu Y, Tao J, Baumgarten KM, Sun H. Hypoxia-mimicking nanofibrous scaffolds promote endogenous bone regeneration. *ACS Appl Mater Interfaces*. 2016;8(47):32450-32459.
33. Yao Q, Sandhurst ES, Liu Y, Sun H. BBP-functionalized biomimetic nanofibrous scaffold can capture BMP2 and promote osteogenic differentiation. *J Mater Chem B*. 2017;5(26):5196-5205.

For article reuse guidelines, please visit Sage's website at http://www.sagepub.com/journals-permissions.